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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: Service Oriented Architecture SOA – (23SDCS03)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AccidentCascade AI: Predicting and Preventing Secondary Road Accidents Using Vehicle Interaction Intelligence	
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Project Architecture / Project Overview:

The proposed **AccidentCascade AI** is a software-based intelligent transportation system designed to detect hazardous traffic situations and predict the possibility of secondary or chain-reaction road accidents using vehicle interaction intelligence. The system takes traffic video data as input and processes the information through a sequence of computer vision, vehicle tracking, feature extraction, interaction analysis, and prediction modules. The primary objective is to move beyond conventional post-accident detection by analyzing vehicle behaviour and estimating the risk of subsequent collisions.

The first stage of the proposed architecture involves vehicle detection using the **YOLOv11 object detection model**. The input traffic video is processed frame by frame to identify relevant road users and vehicles. YOLOv11 provides the detected vehicle information required for the subsequent tracking and analysis stages.

The second stage performs vehicle tracking using **OpenCV**. The detected vehicles are continuously monitored across consecutive frames to obtain their movement information. This stage helps maintain the trajectory of individual vehicles and provides information required to understand changes in their motion over time.

The third stage focuses on **trajectory and feature extraction**. Important vehicle-level characteristics such as position, movement, trajectory, speed-related information, distance and other relevant interaction features are extracted from the tracked data. These features provide the basis for understanding the behaviour of vehicles in different traffic situations.

The fourth stage performs **vehicle interaction analysis**, where the extracted features are used to analyse the relationship between nearby vehicles. Sudden braking, abnormal movement, reduced inter-vehicle distance, lane blockage and other potentially dangerous interactions can be considered for identifying hazardous traffic conditions.

The fifth stage focuses on **accident and hazard detection**. Based on the analyzed vehicle behaviour and interaction patterns, the system identifies potentially dangerous or accident-related situations. Following the identification of a primary accident or hazard, the surrounding vehicle movements are further analyzed to determine whether the event could result in a secondary collision.

The sixth stage introduces secondary accident prediction, which is the primary focus of AccidentCascade AI. The system evaluates the behaviour and trajectories of nearby vehicles to estimate the possibility of a chain reaction following the initial accident. This predictive capability aims to provide information about potential subsequent collisions rather than limiting the system to detecting an accident after it occurs.

The final stages involve the calculation of a **Cascade Risk Score** and generation of an **early warning**. The risk score represents the estimated severity or likelihood of a secondary accident based on the analyzed vehicle interactions. When the predicted risk reaches a critical level, the system can generate an early warning, supporting faster response and potentially reducing the impact of chain-reaction accidents.

Collectively, the proposed architecture integrates **YOLOv11-based vehicle detection, OpenCV-based tracking, trajectory and feature extraction, vehicle interaction analysis, accident/hazard detection, secondary accident prediction, Cascade Risk Scoring, and early warning generation** into a unified software-based framework.

The overall workflow follows the principle of **Detect → Analyze → Predict → Warn → Prevent**, with the aim of improving proactive road-safety management and reducing secondary road accidents.

Project Architechture Flow:

